Thanh-Tai Duong, Ph.D.

Therapeutic Medical Physicist| Iowa-licensed

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Professional Summary

Therapeutic medical physicist with clinical and academic experience since 2009 spanning radiation oncology practice in Vietnam and the United States, university teaching, and postdoctoral research in Canada. Expertise includes Monte Carlo dosimetry, treatment planning, clinical quality assurance, and radiotheranostics. Research focuses on patient-specific radiopharmaceutical therapy dosimetry, quantitative imaging, and independent dose verification. Associate Editor for Medical Physics and Radiation Physics and Chemistry, with experience in student mentoring and multidisciplinary research collaboration.

Licensure and Authorized Medical Physicist Status

- Iowa Medical Physicist License – Radiation Therapy, No. RTH10115 — Valid through April 30, 2027
- Named Authorized Medical Physicist for brachytherapy on Spencer Hospital's radioactive materials license, No. 0164-1-21-M1
- Board certification: Not currently ABR-certified or ABR board-eligible.
- Clinical training and practice obtained outside a conventional U.S. residency pathway.

Core Competencies

- Monte Carlo Simulation & Verification – EGSnrc, TOPAS
- IMRT / VMAT Treatment Planning
- Machine QA, Treatment Planning Verification & Patient-Specific QA
- Radiotheranostics & PET/CT Dosimetry
- Programming – Python, MATLAB
- Deep Learning & Radiomics
- Project Leadership
- Teaching & Mentoring

Working Experience

Jaeger Corporation **Spencer, Iowa, USA**
Therapeutic Medical Physicist, Spencer Hospital *Jan 2025 - Present*

- Perform clinical physics coverage for external beam radiotherapy, including patient-specific QA, physics plan review, weekly chart checks, machine QA, and treatment delivery support.
- Review and support treatment plans for 3D-CRT, IMRT, VMAT, SBRT, and special procedures, with attention to prescription accuracy, dose constraints, image guidance, and treatment delivery safety.
- Generate and evaluate patient-specific QA plans for IMRT/VMAT cases using ArcCHECK and related QA workflows.
- Lead monthly TG-142 machine QA and support routine imaging, mechanical, dosimetric, and safety QA activities.
- Collaborate with radiation oncologists, dosimetrists, therapists, vendors, and engineers to troubleshoot planning, imaging, DICOM transfer, QA, and treatment delivery issues.
- Support treatment planning, record-and-verify, and image-guidance workflows using Monaco, MOSAIQ, and XVI at Spencer Hospital.
- Contribute to SBRT/SRS program development, QA preparation, and end-to-end testing workflows.
- Provide timely physics support for urgent replans, on-treatment issues, and same-day clinical needs.

Western University, Schulich School of Medicine & Dentistry **London, Canada**
Research Associate, Robarts Research Institute *Oct 2024 - present*

Supervisor: Dr. Ting-Yim Lee, Scientist

- Continue collaborative Monte Carlo dosimetry and radiotheranostics projects following the postdoctoral appointment; this is an ongoing research affiliation rather than concurrent primary employment.

Western University, Schulich School of Medicine & Dentistry **London, Canada**
Postdoctoral Associate (PDA), Robarts Research Institute *May 2022 - Oct 2024*

Supervisor: Dr. Ting-Yim Lee, Scientist

- Investigated patient-specific radiopharmaceutical therapy dosimetry using dynamic PET/CT, time-integrated activity estimation, and Monte Carlo simulation.
- Contributed methodology, software, data curation, validation, and manuscript preparation to a six-patient proof-of-concept study of pretreatment dosimetry for ^{177}Lu-PSMA therapy.
- Examined dosimetric and radiobiological modeling of combined radiopharmaceutical therapy and external-beam or brachytherapy treatments; these studies addressed modeled dose rather than demonstrated clinical benefit.

Nguyen Tat Thanh University **Ho Chi Minh City, Vietnam**
Lecturer / Academic Collaborator, Medical Physics *Sep 2022 - present*

- Contribute to medical physics training and student research mentoring through an ongoing academic affiliation.
- Collaborate on medical physics research; this affiliation is distinct from primary clinical employment.

HCMC University of Technology and Education **Ho Chi Minh City, Vietnam**
Lecturer, Department of Industrial Electronics - Biomedical Engineering *Sep 2020 - Sep 2022*

Taught biomedical engineering courses, supervised laboratory work and student projects, developed the Biomedical Imaging Technology curriculum, and contributed to AUN-QA program assessment.

Dong Nai General Hospital **Dong Nai, Vietnam**
Clinical Medical Physicist, Radiation Oncology and Nuclear Medicine *Aug 2009 - Sep 2020*

Provided radiotherapy treatment planning, equipment quality assurance, radiation safety support, clinical research, and teaching.

- Performed linac QA on the Siemens Primus M5497 using TG-142 recommendations and conducted CT simulation and treatment planning system QA.
- Performed 3D-CRT and JO-IMRT treatment planning on Prowess Panther
- Supported clinical implementation and quality assurance of 3D-CRT, IMRT, and HDR brachytherapy.
- Served on the radiation safety committee.

Education

Viet Nam National University-University of Science **Ho Chi Minh City, Vietnam**
Doctor of Philosophy in Nuclear Physics *Sep 2015 – Sep 2019*

- **Advisors:** Prof. Dr. Truong Thi Hong Loan, Dr. rer. nat. Nguyen Dong Son
- **Ph.D. Dissertation:** Dosimetric and Monte Carlo verification of jaws-only IMRT plans calculated by the Collapsed Cone Convolution algorithm for head and neck cancers.
- **GPA: 9.25/10**

Viet Nam National University-University of Science **Ho Chi Minh City, Vietnam**
Master of Science in Nuclear Physics *Sep 2010 – Sep 2012*

- **Advisor:** Prof. Dr. Chau Van Tao
- **Master's Thesis:** Implementing Jaws Only Intensity-Modulated Radiotherapy (IMRT) in Dong Nai Hospital: A Clinical and Technical Evaluation.
- **GPA: 7.5/10** (Thesis Mark: 9.4)

Viet Nam National University-University of Science **Ho Chi Minh City, Vietnam**
Bachelor of Science in Applied Physics *Sep 2005 – Sep 2009*

- **Advisor:** Prof. Dr. Lam Quang Vinh
- **Bachelor's Thesis:** Synthesizing and fabricating CdSe nano quantum dots for cancer theranostics.
- **GPA:** 7.47/10 (Total Accumulated Credits: 210)

Selected publications

- **Duong, T.T.***, De Sarno, D., Fakir, H., Bauman, G., Martinov, M., Thomson, R., and Lee, T.-Y., *A proof-of-concept study of personalized dosimetry for targeted radioligand therapy using pre-treatment diagnostic dynamic PET/CT and Monte Carlo simulation*, **Frontiers in Oncology**, Vol. 15, Article 1600821, **2025**. DOI:10.3389/fonc.2025.1600821.
- **Duong, T.T.***, Nguyen, T.A.T., Tran, T.T., Pham, A.T., Chamberland, M.J.P., Sandwall, P., Bradley, D.A., and Chow, J.C.L., *Novel Validation of HDR Brachytherapy Dosimetry for Cervical Cancer Using* `egs_brachy` *Monte Carlo Simulations: A Comparative Analysis with Oncentra Treatment Planning System*, **Journal of Applied Clinical Medical Physics**, Vol. 26, No. 7, e70144, **July 2025**.DOI:10.1002/acm2.70144.
- **Duong, T.T.***, Luong, T.P., Nguyen, N.A.*, Huynh, V.T.S., Tran, M.L., Nguyen, X.H., Sandwall, P.A., Bradley, D.A. and Chow, J.C.L., *Dosimetric and radiobiological comparison between conventional and hypofractionated breast treatment plans using the Halcyon system*, **Frontiers in Oncology**, Vol. 13, pp.1259416, September **2023**. DOI: 10.3389/fonc.2023.1259416
- **Duong, T.T.***, Luong, T.O., Nguyen, D.S., Truong, T.H.L. and Chow, J.C.L., *Dosimetric and Monte Carlo verification of jaws-only IMRT plans calculated by the Collapsed Cone Convolution algorithm for head and neck cancers*, **Reports of Practical Oncology and Radiotherapy**, Vol. 24, No. 1, pp. 105-114, **2019**. DOI: 10.1016/j.rpor.2018.11.004.
- **Duong, T.T.***, Nguyen, D.S., Truong, T.H.L., and Anson, H.P.W., *Quality assurance of the Jaws Only-Intensity Modulated Radiation Therapy plans for head-and-neck cancer*, **Physica Medica**, Vol. 38, pp. 148-152, **2017**. DOI: 10.1016/j.ejmp.2017.05.059.

Research Experience

Western University, Schulich School of Medicine & Dentistry **London, Canada**
Radiotheranostics project, Robarts Research Institute *May 2022 - present*

Project 1: Monte Carlo (MC) simulation for Radiotheranostics in prostate cancer
- Estimated time-integrated activity from dynamic ^{18}F-DCFPyL PET/CT using Logan graphical analysis and modeled time-activity curves.
- Estimated potential absorbed doses from ^{177}Lu-PSMA-617 using egs_mird, a published Monte Carlo code based on EGSnrc.
- Calculated biologically effective dose (BED) using radiobiological models.
- Developed a patient-specific voxel-based dosimetry workflow linking imaging-derived activity estimates with Monte Carlo calculation.

Project 2: Dosimetric modeling of combined ^{177}Lu-PSMA therapy and external-beam radiotherapy for prostate cancer
- Estimated potential absorbed doses from ^{177}Lu-PSMA-617 using egs_mird
- Calculated the dose for EBRT using Eclipse treatment planning system (TPS)
- Applied radiobiological models to compare BED, tumor control probability (TCP), and normal tissue complication probability (NTCP) for modeled combined treatments.

Project 3: Dosimetric modeling of combined ^{177}Lu-PSMA therapy and HDR/LDR brachytherapy

- Computed the dose for TRT using egs_mird
- Calculated absorbed dose for ^{192}Ir HDR & ^{125}I LDR using egs_brachy, a published Monte Carlo code based on EGSnrc
- Applied radiobiological models to compare BED, tumor control probability (TCP), and normal tissue complication probability (NTCP) for modeled combined treatments.

Project 4: Comparison of Monte Carlo and TG-43 dose calculations for prostate brachytherapy

- Calculated the dose for ^{192}Ir HDR & ^{125}I LDR using egs_brachy
- Compared physical dose distributions and model-derived biological dose metrics between treatment planning system and Monte Carlo calculations.

Dong Nai General Hospital — **Dong Nai, Vietnam**
Department of Radiation Oncology — *2015 – 2020*

Project 1: Calculation of Jaws-only IMRT (JO-IMRT) dose distributions based on the AAPM TG-119 test cases using Monte Carlo simulation and Prowess Panther treatment planning system (TPS).

- Calculated JO-IMRT dose distributions for AAPM TG-119 test cases using Monte Carlo simulation and the Prowess Panther treatment planning system.
- Evaluated multi-target, mock prostate, mock head-and-neck, and C-shape test cases.
- Used independent Monte Carlo calculations to support verification of the Prowess Panther JO-IMRT dose model.
- Implemented DOSCTP-based calculations and compared dose distributions with treatment planning system results.

Project 2: Dosimetric and Monte Carlo verification of Jaws-only IMRT plans calculated by the Collapsed Cone Convolution algorithm for head and neck cancers.

- Independently verified Prowess Panther JO-IMRT dose calculations for head-and-neck treatment plans using Monte Carlo simulation.
- Modeled JO-IMRT beams using BEAMnrc.
- Calculated patient dose distributions for 15 JO-IMRT plans using DOSCTP.
- Compared treatment planning system and Monte Carlo dose distributions to evaluate calculation agreement.

Project 3: Evaluating jaws-only intensity modulated radiation therapy technique at Dong Nai General Hospital by experiment.

- Evaluated JO-IMRT delivery using an IBA FC65-P ionization chamber, MapCHECK 2, and Octavius 4D.
- Assessed jaw-based beam modulation through patient-specific dosimetric measurements.
- Performed measurement-based verification of 25 JO-IMRT plans.
- Used measured dose agreement to assess delivery performance and support clinical implementation.

Research Grants and Funded Projects

- **International Collaborator** — *Deep Learning and Explainable AI–Based Classification of Mammographic Lesions for Early Breast Cancer Detection.* Institut Pertanian Bogor, Indonesia. Riset Kemitraan Internasional, 2026.
- **Co-Investigator** — *Monte Carlo-Based Investigation of Hybrid HDR Brachytherapy and VMAT (HYBIRT) for Tongue Cancer.* Universiti Sains Malaysia, Malaysia. Total project funding: RM 290,000.

Clinical Experience

University Medical ShingMark Hospital **Dong Nai, Vietnam**
Clinical and Research Collaborator, Radiation Oncology *2019 - present*

Since 2019, contributed to development of the radiotherapy program through equipment selection, installation, commissioning, clinical implementation, and research. The continuing affiliation is for research collaboration rather than concurrent primary clinical employment.

- Helped establish a new radiotherapy department.
- Participated in acceptance testing and commissioning of the Varian Halcyon system.
- Trained radiation therapy staff members.
- Developed patient treatment plans for the Halcyon system.
- Collaborated with physician team members

Dong Nai General Hospital **Dong Nai, Vietnam**
Medical Physicist, Department of Radiation Oncology *2009 - 2020*

Combined treatment planning, linac and brachytherapy quality assurance, radiation safety, and clinical research.

I have undertaken the following tasks:

- Performed CT simulation for radiotherapy treatment planning
- Performed external-beam treatment planning for 3D-CRT, IMRT, and VMAT.
- Performed patient-specific QA for IMRT and VMAT using ionization chamber measurements, MapCHECK 2, and Octavius 4D.
- Performed brachytherapy treatment planning and daily QA, and supported source exchange and calibration.
- Operated a Siemens linear accelerator for treatment delivery.
- Performed daily, weekly, and monthly QA on the Siemens Primus M5497 linear accelerator.

Academic Achievements

- **Participation Award by ICTP (Remote):** Awarded a competitive grant to participate in the Joint ICTP–IAEA Workshop on Quantitative Imaging and Analysis Methods in Modern Nuclear Medicine, focused on nuclear imaging quantification, Monte Carlo simulations, and emerging practices (Awarded 2024).
- **Travel Bursary Award by Hokkaido University (Japan) and Stanford University (US):** Funding to join The 4th GI-CoRE/GCB Summer School for Molecular Biomedical Science and Engineering, Japan (Awarded 2022).
- **Travel Bursary Award by Hokkaido University (Japan) and Stanford University (US):** Funding to join the 6th GI-CoRE Summer School for Medical Physics, Japan (Awarded 2019).
- **Travel Bursary Award by International Atomic Energy Agency (IAEA) (Austria):** Funding to join the Regional Training Course on Basic Radiation Dosimetry, Molecular Biology and Radiobiology for Radiotherapy Medical Physics, Malaysia (Awarded 2019).
- **Travel Bursary Award by the Abdus Salam International Centre for Theoretical Physics (ICTP):** Funding to join the Joint ICTP-IAEA Workshop on Monte Carlo Radiation Transport and Associated Data Needs for Medical Application, Italy (Awarded 2017).
- **SEAFOMP Travel Awards (Thailand):** Funding to present PhD's thesis work at the 22nd International Conference on Medical Physics, Thailand (Awarded 2016).
- **Travel Bursary Award by the Abdus Salam International Centre for Theoretical Physics (ICTP) (Italy):** Funding to join the School on Medical Physics for Radiation Therapy: Dosimetry and Treatment Planning for Basic and Advanced Applications, Italy (Awarded 2015).
- **Student Research Award (Vietnam):** Support for an outstanding student research project at VNUHCM-University of Science, Vietnam (2008).
- **Student Research Award (Vietnam):** Recognition and support for outstanding student achievement at VNUHCM-University of Science, Vietnam (2007, 2008).

Editorial Activities

Editorial Roles

- Associate Editor, *Medical Physics* (AAPM/Wiley) — Dec 2025–Dec 2028
- Associate Editor, *Radiation Physics and Chemistry* (Elsevier) — 2025–Present
- Editorial Board Member, *Biomedical Physics & Engineering Express* (IOP Publishing)
- Executive Guest Editor, *Clinical Integration and Innovation with the Halcyon/Ethos™ Radiotherapy System: Global Perspectives and Practical Experiences*, *Journal of Radiation Research and Applied Sciences* (Elsevier) — 2025–2026

Reviewer for Scientific Journals
(26 journals in total)

- Medical Physics
- Physics in Medicine & Biology
- Computer Methods and Programs in Biomedicine
- Journal of Applied Clinical Medical Physics
- Physica Medica: European Journal of Medical Physics
- Physics and Imaging in Radiation Oncology
- Applied Radiation and Isotopes
- Radiation Physics and Chemistry
- Nuclear Engineering and Technology
- Nuclear Science and Engineering
- Acta Neurologica Scandinavica
- Biomedical Signal Processing and Control
- Journal of Medical Imaging and Radiation Sciences
- Journal of Medical Physics
- Reports of Practical Oncology and Radiotherapy
- Radiation and Environmental Biophysics
- Technology in Cancer Research & Treatment
- Cancer Management and Research
- Biomedical Physics & Engineering Express
- Journal of Cancer Research and Therapeutics
- Journal of X-Ray Science and Technology
- Science Progress
- Cancer Reports
- The Egyptian Journal of Radiology and Nuclear Medicine
- Array
- Atom Indonesia

Featured in Media Coverage

- **Thanh Niên Newspaper**, Sep 5, 2025: "Sharing the same bloodline of Lac Hong Aspiration to improve the quality of cancer treatment in Vietnam" [Vietnamese] [Translated]
- **Dong Nai Newspaper**, Jul 29, 2025: "Personalizing cancer treatment with targeted radiation therapy" [Vietnamese] [Translated]
- **VietnamNet Newspaper**, Feb 19, 2025: "The surprising decision of a U40 doctor in Dong Nai" [Vietnamese] [Translated]
- **Dong Nai Newspaper**, Jan 24, 2025: "Journey from Dong Nai General Hospital to Harvard University." [Vietnamese] [Translated]
- **Dong Nai Newspaper**, Nov 13, 2016: "Medical physicist profession" [Vietnamese] [Translated]

Skills

Programming	Python, MATLAB
Monte Carlo	EGSnrc, TOPAS, GAMOS
Software	LIFEx, 3D Slicer, ITK, CERR, DOSCTP, SPSS, Origin
Treatment planning	Monaco, Eclipse, Prowess Panther, BrachyVision
Oncology information systems	MOSAIQ, ARIA

Languages

Vietnamese	Mother tongue.
English	Professional use in clinical practice and research.

Publications

Summary: Publications in medical physics, radiation oncology, Monte Carlo dosimetry, brachytherapy, radiotheranostics, imaging, and related fields. The list includes journal articles, published conference abstracts, and proceedings papers; abstracts and proceedings are labeled. An asterisk denotes corresponding authorship. Selected publications above are repeated here for completeness.

[67] **Duong, T.T.***, Tran, D., Bradley, D.A., and Chow, J.C.L., *Clinical applications of the Halcyon O-ring linear accelerator in radiation therapy: a systematic review of treatment efficacy, workflow efficiency, and imaging integration*, **Physica Medica**, Vol. 150, Article 107206, **2026**. DOI:10.1016/j.ejmp.2026.107206.

[66] [Proceedings paper] Du, N.M.*, Ngo, H.H., and **Duong, T.T.**, *Optimization of proton energy and target thickness for cyclotron-based ^{18}F production: An ISOTOPIA simulation study*, **Journal of Physics: Conference Series**, Vol. 3284, Article 012027, **2026**. DOI:10.1088/1742-6596/3284/1/012027.

[65] Yani, S.*, Nikmah, K., Kholis, T.S.N., Ramadhan, M.V., Sumaryada, T., Fitrianingsih, E., Nasser, E.N., Pratomo, B., Budiantoro, P.A., Mayditia, H., **Duong, T.T.***, Tamam, N., and Sulieman, A., *Evaluation of multi-layer shielding effectiveness against high-energy protons and secondary radiation in LEO Satellites using PHITS*, **Radiation Physics and Chemistry**, Vol. 248, Article 114076, **2026**. DOI:10.1016/j.radphyschem.2026.114076.

[64] Yani, S., Rahma, W., Sumaryada, T., Endarko, Haryanto, F., **Duong, T.T.***, Omer, H., Tamam, N., Sulieman, A., Dirgayussa, I.G.E., and Nuraeni, N., *Optimization of proton radiation shielding using multilayer concrete-iron-polyethylene barriers: A PHITS Monte Carlo study*, **Radiation Physics and Chemistry**, Vol. 240, Article 113422, **2026**. DOI:10.1016/j.radphyschem.2025.113422.

[63] Tran, D., **Duong, T.T.***, Wadi-Ramahi, S., Bradley, D., and Chow, J.C.L., *Beyond TG-43: A PRISMA-based systematic review on model-based dose-calculation algorithms in brachytherapy*, **Journal of Applied Clinical Medical Physics**, Vol. 27, Article e70627, **2026**. DOI:10.1002/acm2.70627.

[62] **Duong, T.T.***, Luong, T.P., Nguyen, T.L., Nguyen, V.T., Ngo, H.L., Dang, Q.H., Mac, D.Q., Nguyen, N.A., and Chow, J.C.L., *Rectal cancer radiotherapy planning in Vietnam: A multi-center dosimetric and radiobiological study*, **Radiation Physics and Chemistry**, Vol. 244, Article 113780, **2026**. DOI:10.1016/j.radphyschem.2026.113780.

[61] **Duong, T.T.***, Luong, T.P., Tran, T.K., Duong, T.L., Nguyen, N.A., Nguyen, Q.H., Sandwall, P., Alaei, P., Bradley, D., and Chow, J.C.L., *From standard to stratified: Modeling NTCP and EAR to personalize daily MV-CBCT in radiotherapy*, **Journal of Applied Clinical Medical Physics**, Vol. 26, Article e70414, **2025**. DOI:10.1002/acm2.70414.

[60] Thach, L.B., Thao, M.T., Nghia, N.V., My, T.N., **Duong, T.T.***, Tamam, N., Sulieman, A., Omer, H., Toufig, H., and Bradley, D., *Hippocampal-Avoidance Whole-Brain Radiotherapy: Dosimetric Comparison of 3D-CRT, IMRT, and VMAT for Brain Metastases from Lung Cancer*, **Cancers**, Vol. 17, Article 3744, **2025**. DOI:10.3390/cancers17233744.

[59] Alfajri, A.I., Sumaryada, T., Yani, S.*, **Duong, T.T.***, Tamam, N., and Sulieman, A., *Hybrid quantum-classical convolutional neural networks for multiclass lung X-ray image classification*, **Journal of Radiation Research and Applied Sciences**, Vol. 19, Article 102165, **2026**. DOI:10.1016/j.jrras.2026.102165.

[58] Ahmad, I.M., Husin, A.D., **Duong, T.T.***, Tamam, N., Sulieman, A., and Yani, S.*, *Evaluation of the Lawson criterion for aneutronic proton-boron-11 fusion: Effects of ion temperature and bremsstrahlung losses*, **Frontiers in Nuclear Engineering**, Vol. 5, Article 1714531, **2026**. DOI:10.3389/fnuen.2026.1714531.

[57] Sulieman, A., Almogren, K.S.*, Siddig, E., Salah, H., Alrebdi, H., Tamam, N., Alkhallas, F., Abuhadi, N., Omer, H., **Duong, T.T.**, and Bradley, D.A., *Assessment of occupational exposure and safety for customs and border protection personnel in Saudi Arabia*, **Radiation Physics and Chemistry**, Vol. 245, Article 113887, **2026**. DOI:10.1016/j.radphyschem.2026.113887.

[56] Yani, S., Nugraha, S.P., Alfajri, A.I., Sumaryada, T., Duong, T.T.*, Omer, H., Tamam, N., and Sulieman, A., Quantum convolutional neural network for MRI brain tumor analysis: Binary and multiclass classification using a four-qubit architecture, Radiation Physics and Chemistry, Vol. 242, Article 113545, 2026. DOI:10.1016/j.radphyschem.2025.113545.

[55] **Duong, T.T.***, De Sarno, D., Fakir, H., Bauman, G., Martinov, M., Thomson, R., and Lee, T.-Y., *A proof-of-concept study of personalized dosimetry for targeted radioligand therapy using pre-treatment diagnostic dynamic PET/CT and Monte Carlo simulation*, **Frontiers in Oncology**, Vol. 15, Article 1600821, **2025**. DOI:10.3389/fonc.2025.1600821.

[54] **Duong, T.T.***, Nguyen, T.A.T., Tran, T.T., Pham, A.T., Chamberland, M.J.P., Sandwall, P., Bradley, D.A., and Chow, J.C.L., *Novel Validation of HDR Brachytherapy Dosimetry for Cervical Cancer Using* `egs_brachy` *Monte Carlo Simulations: A Comparative Analysis with Oncentra Treatment Planning System*, **Journal of Applied Clinical Medical Physics**, Vol. 26, No. 7, e70144, **July 2025**.DOI:10.1002/acm2.70144.

[53] Pham, N.T., Phan, C.D., Nguyen, V.T., Nguyen, L.T., Thao, M.T., **Duong, T.T.***, Sandwall, P., Ashmeg, S., Sulieman, A., and Chow, J.C.L., *Application of Volumetric Modulated Arc Therapy (VMAT) for Craniospinal Irradiation: Dosimetric Analysis and Clinical Implementation*, **Radiation Physics and Chemistry**, Vol. 236, Article 112965, **2025**. DOI: 10.1016/j.radphyschem.2025.112965.

[52] **Duong, T.T.***, Loc, T.M., Kien, T.T., Phat, L.T., Anh, N.N., Omer, H., Sulieman, A., Bradley, D.A., and Chow, J.C.L., *Assessing efficacy and toxicity in rectal cancer radiotherapy: a comparative study of two fractionation regimens*, **Oncologie**, Vol. 27, No. 4, pp. 509-518, **2025**. DOI: 10.1515/oncologie-2025-0037.

[51] Yani, S., Kholis, T.S.N., **Duong, T.T.***, Tamam, N., Sulieman, A., Sumaryada, T., Akhiruddin, A., Wahyudi, S.T., and Arif, I.,*Assessment of Proton Shielding Properties of Rare Earth Metal-Concrete Using PHITS Monte Carlo Code*,**Radiation Physics and Chemistry**, Vol. 236, 112908, **November 2025**. DOI:10.1016/j.radphyschem.2025.112908.

[50] **Duong, T.T.**, Nguyen, N.A., Phat, L.T., Kien, T.T., Hai, N.X., Tiep, N.H., Sandwall, P., Omer, H., Sulieman, A., Tamam, N., and Chow, J.C.L.,*Impact of Radiobiological Parameters on Tumor Control Probability in Rectal Cancer Using the Poisson Linear Quadratic Model*,**Radiation Physics and Chemistry**, Vol. 236, 112902, **November 2025**.DOI:10.1016/j.radphyschem.2025.112902.

[49] **Duong, T.T.***, Phat, L.T., Kien, T.T., Nguyen, N.A., Hai, N.X., Sandwall, P., Bradley, D., and Chow, J.C.L.,*Enhancing Patient Positioning Accuracy: Evaluating Daily Cone Beam Computed Tomography in the Halcyon System*,**Japanese Journal of Clinical Oncology**, Vol. 55, No. 8, pp. 925-932, **2025**. DOI: 10.1093/jjco/hyaf067.

[48] Thao, M.T., Thach, L.B., Phat, L.T., Thanh, N.V., **Duong, T.T.***, Sandwall, P., Sulieman, A., Tamam, N., and Chow, J.C.L., *Dosimetric Validation of Elekta Synergy 6 MV and 10 MV Photon Beam Models in the Monaco*, **Radiation Physics and Chemistry**, Vol. 236, Article 112819, **2025**. DOI: 10.1016/j.radphyschem.2025.112819.

[47] Nguyen, V.T., Pham, N.T., **Duong, T.T.***, Truong, T.H.L., Sandwall, P., Bradley, D.A., and Chow, J.C.L., *A Comparative Dosimetric Evaluation of Dynamic Conformal Arc Therapy and Volumetric Modulated Arc Therapy for Lung Stereotactic Body Radiotherapy*, **Nuclear Engineering and Technology**, Vol. 57, No. 4, Article 103311, **2025**. DOI: 10.1016/j.net.2024.11.013.

[46] **Duong, T.T.***, Pham, N.T., Hoang, D.T., Hoang, T.P.H., Kandemir, R., Omer, H., Sulieman, A., Bradley, D.A., and Chow, J.C.L.,*Scanning Protocol Influence on Relative Electron Density–CT Number Calibrations and Radiotherapy Dose Calculation for a Halcyon Linac*,**Radiation Physics and Chemistry**, Vol. 236, 112760, **November 2025**.DOI:10.1016/j.radphyschem.2025.112760.

[45] Phat, L.T., Thao, M.T., Kien, T.T., **Duong, T.T.***, Sandwall, P., Sulieman, A., Tamam, N., Yani, S., and Chow, J.C.L., *Dosimetric Analysis of Beam Number Variations in IMRT for Head-and-Neck, Breast, and Pelvic Cancers Using Halcyon*, **Radiation Physics and Chemistry**, Vol. 236, Article 112755, **2025**. DOI: 10.1016/j.radphyschem.2025.112755.

[44] Nguyen, K.S., Luong, T.P., Ha, A.T., Ngo, H.L., **Duong, T.T.***, Booker, P., Yani, S., Sulieman, A., Tamam, N., Sandwall, P., and Chow, J.C.L., *Evaluation of Multi-Leaf Collimator Accuracy in Halcyon System Using Logfile Analysis*, **Radiation Physics and Chemistry**, Vol. 236, Article 112741, **2025**. DOI: 10.1016/j.radphyschem.2025.112741.

[43] Nguyen, K.S., Ngo, H.L., Ha, A.T., Lam, B., **Duong, T.T.***, Tamam, N., Sulieman, A., Sandwall, P., and Chow, J.C.L., *Improving Radiotherapy Precision Through MLC Model Optimization for a Monaco Treatment Planning System*, **Radiation Physics and Chemistry**, Vol. 236, Article 112730, **2025**. DOI: 10.1016/j.radphyschem.2025.112730.

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[41] **Duong, T.T.***, Hoang, D.T., Nguyen, X.H., Nguyen, N.A., Bradley, D.A., Sandwall, P.A., and Chow, J.C.L.,*Examining Beam Matching in the Commissioning of a Halcyon Accelerator*,**Nuclear Science and Engineering**, Vol. 199, No. 2, pp. 314-324, **2025**. DOI: 10.1080/00295639.2024.2366733.

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Teaching Experience

- HCMC University of Technology and Education, Ho Chi Minh, Viet Nam
  - Biomedical Imaging Technology (BIIM332865)- Average number of students annually: 50
  - Biomedical Image Processing Lab (BIMP311865) - Average number of students annually: 25
  - Biomedical Information Systems Lab (BSPR411965) - Average number of students annually: 25
  - Capstone Project (CAPR411265) - Average number of students annually: 6
- Lac Hong University, Dong Nai, Viet Nam
  - General Physics Laboratory for Pharmacy (Phys102012) - Average number of students annually: 100
- Tra Vinh University, Tra Vinh, Viet Nam and Dong Nai University, Dong Nai, Viet Nam
  - Biophysics (BIO04213) - Average number of students annually: 100

Mentoring Experience

Summary: I have supervised over 25 undergraduate students on their thesis projects. Here are a few examples.

- HCMC University of Technology and Education, Ho Chi Minh, Viet Nam
  - Student's Name: Vo Kieu Phuong
    - Period: 2017-2021
    - Project: Application of deep learning and radiomics to lung cancer prediction
    - Current Position: Product Specialist, SAGOMED JSC
  - Student's Name: Nguyen Huynh Tai
    - Period: 2017-2021
    - Project: The initial design and construction of a 12 Channel Electrocar-diogram Device developed on an ADS1293 chip platform
    - Current Position: Technical Staff, Hoang Le Medical Equipment Co.,Ltd

- University of Science - VNUHCM, Ho Chi Minh, Viet Nam
  - Student's Name: Tran Phat
    - Period: 2012-2016
    - Project: Patient specific quality assurance for IMRT using MatrixX and Octavius 4D
    - Current Position: Medical physicist, Children's Hospital
  - Student's Name: Hoang Duc Tuan
    - Period: 2012-2016
    - Project: Monte Carlo simulation for LINAC
    - Current Position: Medical physicist, Shingmark hospital
  - Student's Name: Nguyen Thi Hong Trang
    - Period: 2012-2016
    - Project: Dosimetric and radiobiological comparison between 3D-CRT and JO-IMRT
    - Current Position: Medical physicist, HCMC Oncology Hospital
  - Student's Name: Luong Thi Oanh
    - Period: 2013-2017
    - Project: Dosimetric evaluation and Monte Carlo simulation of AAPM TG-119 test cases
    - Current Position: Medical physicist, HCMC Oncology Hospital
  - Student's Name: Nguyen Thi Hoa
    - Period: 2014-2018
    - Project: Evaluation of IMRT using radiobiological models
    - Current Position: Medical physicist, HCMC Oncology Hospital
- Nguyen Tat Thanh University, Ho Chi Minh, Viet Nam
  - Student's Name: Nguyen Truong Sang
    - Period: 2018-2022
    - Project: Application of deep learning and radiomics to head-and-neck cancer prediction
    - Current Position: Medical physicist, Cho Ray Hospital

Memberships in Professional Societies

- American Association of Physicists in Medicine (AAPM): 2012-present
- Canadian Organization of Medical Physicists (COMP): March 2024-present
- Vietnam Society of Medical Physics (VSMP): 2009-present
- International Radiation Physics Society (IRPS): 2022-present
- Vietnam Physical Society (VPS): 2009 – present
- European Federation of Organisations for Medical Physics (EFOMP): July, 2024 – present

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